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BSCS / C303

Finals Essay Task 1. Reading Task on Analytics

Instruction:

Read the attached Article and video link before answering the questions.

Video link: <https://youtu.be/hGCKdMsfU7g>

Article Link:

<https://drive.google.com/file/d/1FhanZgD5YFiBJP0088UbEWaN7DEFbgiE/view?usp=sharing>

Following the video that you have watched and article that I ask you to read please answer the following question using your own words in 2–3 sentences based on the provided source material.

1. What are the four primary factors that will differentiate a data analyst candidate in 2026?

- The four main factors that will make a data analyst stand out in 2026 are domain knowledge, understanding the technical ecosystem, strong communication skills, and having a clear reason or motivation. These qualities matter because companies now want analysts who can solve business problems, work with teams, and stay consistent in their career growth.

2. How does domain knowledge provide a competitive advantage over purely technical candidates?

- Domain knowledge gives a candidate an advantage because they already understand how a specific industry works, including its metrics, customers, and common challenges. This makes them more valuable than someone who only has technical skills but no understanding of the business side.

3. What is the "lean tech stack" recommended for a foundational data analyst role?

- The recommended lean tech stack for beginners includes Excel, SQL, and one visualization tool such as Tableau, Power BI, or Looker. These tools are enough to build a strong foundation because they cover data cleaning, querying, analysis, and presenting insights.

4. Distinguish between "learning projects" and "showing projects."

- Learning projects are simple practice projects that focus on improving one tool at a time using clean datasets. Showing projects are more advanced projects that solve real business questions using messy data, multiple tools, and clear storytelling to impress employers.

5. According to the roadmap, why is Python often deprioritized for mid-career data analyst positions?

- Python is often deprioritized for mid-career data analyst roles because it is not commonly tested in interviews and is less used in daily business-facing tasks. Employers usually prioritize tools like SQL, dashboards, and business communication since stakeholders can understand these outputs more easily.

Questions on SQL-Base on the Article Posted

6. Why is establishing the "Business Context" essential for an analyst? What is a Business Context?

- Establishing the business context is essential because it helps analysts understand the purpose of the data and the problem that needs to be solved. Business context means knowing what each row represents, what question the data should answer, and who will use the results.

7. According to the text, how does SQL function as a "flashlight"?

- SQL functions like a flashlight because it helps users explore data and uncover hidden issues or patterns. It allows analysts to spot errors such as wrong ages, invalid dates, duplicate records, and unusual values.

8. Why is the validation step considered a professional necessity rather than an optional task?

- Validation is considered necessary because it prevents mistakes and ensures the data is accurate before analysis begins. It also helps build trust and credibility since wrong data can lead to incorrect conclusions.

9. What defines "good SQL" according to the author's standards?

- Good SQL is defined as queries that are easy to read, easy to explain, and easy to trust. It focuses more on clarity and usefulness rather than making long or complicated queries.

10. Describe the fundamental mindset shift required to move from being a student to being an analyst.

- The mindset shift is moving from simply writing queries to using queries to answer business questions. Instead of focusing only on technical tasks, an analyst thinks about how data can solve real problems and support decisions.